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Minimum viable charge exchange for service operations in edge devices

Abstract

A computing device determines that an action should be performed on a first edge device operating in a computing system. The computing device accesses a data structure that identifies values of computing resources of the first edge device and computational requirements of performing the action on the first edge device. The computing device determines, based on the data structure, that the first edge device lacks a computing resource to meet the computational requirements of performing the action on the first edge device. In response to determining that the first edge device lacks the computing resource to meet the computational requirements of performing the action on the first edge device, the computing device inhibits the action from being performed on the first edge device.

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Background/Summary

BACKGROUND

(1) Edge devices are computing devices in a computing system or network that are located at the edge of the computing system or network. Edge devices may be low power and require adaptive charge sharing in order to perform certain operations, however, adaptive charge sharing may not correlate to the intended operation, leaving an edge device inoperable.

SUMMARY

(2) The examples disclosed herein implement a charge exchange service that performs minimum

viable charge exchange for service operations in edge devices. In particular, the charge exchange service can correlate the time an operation or action takes to occur and the battery expenditure required to power the edge device for the duration of the operation or action. The charge exchange service can identify the minimum viable battery charge for the edge device and determine whether the battery expenditure required to perform the intended operation or action will leave the edge device with the minimum viable battery charge for the edge device.

(3) In one example, a method for minimum viable charge exchange for service operations in edge devices is provided. The method includes determining, by a computing device, that an action should be performed on a first edge device operating in a computing system. The method further includes accessing, by the computing device, a data structure that identifies values of computing resources of the first edge device and computational requirements of performing the action on the first edge device. The method further includes determining, by the computing device based on the data structure, that the first edge device lacks a computing resource to meet the computational requirement of performing the action on the first edge device. The method further includes, in response to determining that the first edge device lacks the computing resource to meet the computational requirement of performing the action on the first edge device, inhibiting, by the computing device, the action from being performed on the first edge device.

(4) In another example, a computing device for minimum viable charge exchange for service operations in edge devices is provided. The computing device includes a memory and a processor device coupled to the memory. The processor device is to determine that an action should be performed on a first edge device operating in a computing system. The processor device is further to access a data structure that identifies values of computing resources of the first edge device and computational requirements of performing the action on the first edge device. The processor device is further to determine, based on the data structure, that the first edge device lacks a computing resource to meet the computational requirement of performing the action on the first edge device. The processor device is further to, in response to determine that the first edge device lacks the computing resource to meet the computational requirement of performing the action on the first edge device, inhibit the action from being performed on the first edge device.

(5) In another example, a non-transitory computer-readable storage medium is provided. The non-transitory computer-readable storage medium includes computer-executable instructions to cause a processor device to determine that an action should be performed on a first edge device operating in a computing system. The instructions further cause the processor device to access a data structure that identifies values of computing resources of the first edge device and computational requirements of performing the action on the first edge device. The instructions further cause the processor device to determine, based on the data structure, that the first edge device lacks a computing resource to meet the computational requirement of performing the action on the first edge device. The instructions further cause the processor device to, in response to determine that the first edge device lacks the computing resource to meet the computational requirement of performing the action on the first edge device, inhibit the action from being performed on the first edge device.

(6) Individuals will appreciate the scope of the disclosure and realize additional aspects thereof after reading the following detailed description of the examples in association with the accompanying drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawing figures incorporated in and forming a part of this specification illustrate several aspects of the disclosure and, together with the description, serve to explain the

principles of the disclosure.

- (2) FIG. 1 is a block diagram of a computing device in which examples of minimum viable charge exchange for service operations in edge devices may be practiced;
- (3) FIG. 2 is a flowchart illustrating operations performed by the computing device of FIG. 1 for minimum viable charge exchange for service operations in edge devices, according to one example;
- (4) FIG. 3 is a block diagram of the computing device of FIG. 1 for minimum viable charge exchange for service operations in edge devices, according to one example;
- (5) FIG. 4 is a block diagram of the computing device of FIG. 1 for minimum viable charge exchange for service operations in edge devices, according to one example;
- (6) FIG. 5 is a block diagram of the computing device of FIG. 1 for minimum viable charge exchange for service operations in edge devices, according to one example;
- (7) FIG. 6 is a block diagram of the computing device of FIG. 1 for minimum viable charge exchange for service operations in edge devices, according to one example;
- (8) FIG. 7 is a block diagram of the computing device of FIG. 1 for minimum viable charge exchange for service operations in edge devices, according to one example;
- (9) FIG. 8 is a block diagram of the computing device of FIG. 1 for minimum viable charge exchange for service operations in edge devices, according to one example; and
- (10) FIG. 9 is a block diagram of a computing device suitable for implementing examples, according to one example.

DETAILED DESCRIPTION

(11) The examples set forth below represent the information to enable individuals to practice the examples and illustrate the best mode of practicing the examples. Upon reading the following description in light of the accompanying drawing figures, individuals will understand the concepts of the disclosure and will recognize applications of these concepts not particularly addressed herein. It should be understood that these concepts and applications fall within the scope of the disclosure and the accompanying claims.

(12) Any flowcharts discussed herein are necessarily discussed in some sequence for purposes of illustration, but unless otherwise explicitly indicated, the examples are not limited to any particular sequence of steps. The use herein of ordinals in conjunction with an element is solely for distinguishing what might otherwise be similar or identical labels, such as “first message” and “second message,” and does not imply an initial occurrence, a quantity, a priority, a type, an importance, or other attribute, unless otherwise stated herein. The term “about” used herein in conjunction with a numeric value means any value that is within a range of ten percent greater than or ten percent less than the numeric value. As used herein and in the claims, the articles “a” and “an” in reference to an element refers to “one or more” of the elements unless otherwise explicitly specified. The word “or” as used herein and in the claims is inclusive unless contextually impossible. As an example, the recitation of A or B means A, or B, or both A and B. The word “data” may be used herein in the singular or plural depending on the context.

(13) Edge devices are computing devices that operate at the edge of a computing system or network, near a user or data source rather than a centralized data center or cloud. Edge devices may be Internet of Things (IOT) devices or gateways, computer networking devices, automobiles, smart devices, smart displays, or any computing device that is located at the edge of a network and contains hardware, such as memory, a processor, storage, and computing resources, to run services and collect and process data. Edge devices may also be referred to as fog devices, and edge devices and fog devices may be components of edge computing. It is to be understood that the examples herein are not limited to edge devices and an edge device may be a fog device or any computing device that operates at the edge of a computing system or network near a user or data source.

(14) Edge devices in a computing system or network may be characterized as being low power. As a result, edge devices require a power source or other form of charge or power in order to operate in a computing system or network. In some cases, an action, such as updating or installing a

component on an edge device, may result in reducing the power level of the edge device to a level below a minimum viable charge for the edge device, where the edge device can no longer operate in the computing system or network. In some cases, edge devices may require adaptive charge sharing with other edge devices in the computing system or network in order to complete the action. However, if an incorrect amount of power is transferred, then the edge device transferring power may need more battery charge to be able to stay out in the field and the edge device receiving power may not receive enough power to successfully complete the action. These power exchanges do not correlate to the intended action, causing inefficiencies in the edge devices that comprise a computing system or network.

(15) When an edge device intends to perform an action that requires battery power, such as installing or updating a component of the edge device, a charge exchange service can determine whether the action will consume an amount of battery power in excess of the minimum viable charge for the edge device that allows the edge device to remain operational after the action is completed. As a result, the edge device can perform the action if the charge exchange service determines that the action will consume an amount of battery that leaves the edge device with at least the minimum viable charge for the edge device, and the edge device cannot perform the action if the charge exchange service determines that the action will consume an amount of battery that will leave the edge device with less than the minimum viable charge for the edge device.

(16) The charge exchange service can determine whether the action will consume an amount of another computing resource of the edge device, such as memory or storage, that will render the edge device inoperable. As a result, the edge device can perform the action if the charge exchange service determines that the action will consume an amount of a computing resource that leaves the edge device with an amount of the computing resource that allows the edge device to operate, or the edge device cannot perform the action if the charge exchange service determines that the action will consume an amount of the computing resource that will leave the edge device with less than the amount of the computing resource that will allow the edge device to operate.

(17) In order to determine whether the action will consume an amount of battery power in excess of the minimum viable charge for the edge device, or in excess of an amount of another computing resource of the edge device that will render the edge device inoperable, the charge exchange service can access a data structure, such as a table, that identifies the edge devices in the computing system or network, the values of the computing resources of the edge devices, and the computational requirements, such as battery power required, for actions that can be performed on the edge device. The values of the computing resources of the edge devices and the computational requirements for actions that can be performed on the edge device can be determined by testing and running the edge device in production, monitoring the battery and CPU cycles of the edge device, taking key measurements of computing resources around the action, and in other manners that allow for measuring the computing resources of edge devices and actions that can be performed on edge devices. The data structure can be updated over time as edge devices age and computational task intensity changes, with the hardware profile being generated and made available, allowing for better design and strategies around updates.

(18) The charge exchange service can also transfer battery power between edge devices in the computing system or network when an edge device in the computing system or network does not have enough battery power to perform the action and another edge device in the computing system or network has an amount of battery power in excess of the amount of battery power needed to operate in the computing system or network. Based on the information in the data structure, the edge device with an amount of battery power in excess of the amount of battery power needed to operate can share battery with the edge device that does not have enough battery power to perform the action, in an amount that will allow both edge devices to operate.

(19) FIG. 1 is a block diagram of a computing device **10** that comprises a system memory **12**, a processor device **14**, and a storage device **16** in which examples of minimum viable charge

exchange for service operations in edge devices may be practiced. It is to be understood that the computing device **10** in some examples may include constituent elements in addition to those illustrated in FIG. **1**. In the example of FIG. **1**, the computing device **10** implements a charge exchange service **18** that performs minimum viable charge exchange for service operations in edge devices.

(20) The charge exchange service **18** may determine that an action **20** should be performed on a first edge device **22-1** that is operating in a computing system or network. In some examples, the action **20** may be to update an application **24**, update firmware **26**, update the operating system **28**, update a package **30**, update a service **32**, install a service, **34**, update software **36**, or install software **38** on the first edge device **22-1**, as non-limiting examples.

(21) The charge exchange service **18** can access a data structure **40** that identifies computing resource values **42** of the first edge device **22-1** and computational requirements **44** of performing the action **20** on the first edge device **22-1**. The data structure **40** may be any data structure that contains data about the computing resource values **42** of the first edge device **22-1** and the computational requirements **44** of performing the action **20** on the first edge device **22-1**, such as a table, array, record, hash map, dictionary, list, linked list, graph, or tree, as non-limiting examples.

(22) The computing resource values **42** of the first edge device **22-1** may identify quantities of computing resources of the first edge device **22-1**. The computing resource values **42** of the first edge device **22-1** may contain multiple entries that correlate to the quantities of multiple computing resources of the first edge device **22-1**. For example, the computing resource values **42** of the first edge device **22-1** may contain a quantity of a computing resource **46** of the first edge device **22-1** and a quantity of a computing resource **48** of the first edge device **22-1**. The quantity of a computing resource **46** of the first edge device **22-1** may correlate to a computing resource **50** of the first edge device **22-1** and the quantity of a computing resource **48** of the first edge device **22-1** may correlate to a computing resource **52** of the first edge device **22-1**. A computing resource (e.g., the computing resource **50**, the computing resource **52**) of the first edge device **22-1** may be one or more of CPU cycles **54**, available memory **56**, available storage **58**, load **60**, battery level **62**, or age **64** of the first edge device **22-1**, as non-limiting examples, the quantities of which can be represented in the computing resource values **42** of the first edge device **22-1** in the data structure **40**. For example, the computing resource **50** of the first edge device **22-1** may be the battery level **62** of the first edge device **22-1** with a value of 20%, which can be signified in the quantity of a computing resource **46** of the first edge device **22-1** indicated in the computing resource values **42** of the first edge device **22-1** of the data structure **40**. In another example, the computing resource **50** of the first edge device **22-1** may be the battery level **62** of the first edge device **22-1** with a value of 20%, which can be indicated in the quantity of a computing resource **46** of the first edge device **22-1** of the computing resource values **42** of the first edge device **22-1**, and the computing resource **52** of the first edge device **22-1** may be the available memory **56** of the first edge device **22-1** with a value of 2 GB, which can be indicated in the quantity of a computing resource **48** of the first edge device **22-1** of the computing resource values **42** of the first edge device **22-1**.

(23) The computational requirements **44** of performing the action **20** on the first edge device **22-1** may be one or more of CPU cycles **66**, memory **68**, storage **70**, network capabilities **72**, battery **74**, or time **76** required to perform the action **20** on the first edge device **22-1**, as non-limiting examples. For example, the action **20** to perform on the first edge device **22-1** may be to update software **36** and the computational requirements **44** of performing the action **20** on the first edge device **22-1** may indicate that in order to update software **36**, the memory **68** required is 1 GB, the storage **70** required is 5 GB, and the battery **74** required is 10%.

(24) The data structure **40** may identify the values of computing resources of a plurality of edge devices operating in the computing system or the network. Computing resource values **78** of an edge device from among the plurality of edge devices may be identified in the data structure **40**. The computing resource values **78** of an edge device from among the plurality of edge devices can

indicate the quantities of computing resources of an edge device from among the plurality of edge devices. For example, computing resource values **78** of an edge device may identify quantities of computing resources of the edge device, such as one or more of CPU cycles, available memory, available storage, load, battery level, and age of the edge device, as non-limiting examples. The data structure **40** may contain an entry (e.g., the computing resource values **78** of an edge device) for each edge device in the plurality of edge devices that identifies the computing resource values for the edge device in the form of the quantities of the computing resources of the edge device.

(25) The data structure **40** may identify the computational requirements of performing actions on the plurality of edge devices operating in the computing system or the network. Computational requirements **80** of performing an action on an edge device from among the plurality of edge devices may be identified in the data structure **40**. The computational requirements **80** of performing an action on an edge device from among the plurality of edge devices can indicate one or more of the CPU cycles, memory, storage, network capabilities, battery, or time required to perform the action, such as updating an application, updating firmware, updating the operating system, updating a package, updating a service, installing a service, updating software, or installing software on the edge device from among the plurality of edge devices, as non-limiting examples. The data structure **40** may contain an entry (e.g., the computational requirements **80** of performing an action) for each action (e.g., the action **20**) that can be performed on each edge device from among the plurality of edge devices.

(26) The charge exchange service **18** may determine, based on the data structure **40**, that the first edge device **22-1** lacks a computing resource (e.g., the computing resource **50**) sufficient to meet the computational requirements **44** of performing the action **20** on the first edge device **22-1**. The charge exchange service **18** may then inhibit the action **20** from being performed on the first edge device **22-1**. For example, the computing resource **50** of the first edge device **22-1** may be the battery level **62** of the first edge device **22-1** with a value of 20%, which may be identified in the quantity of a computing resource **46** of the first edge device **22-1** in the computing resource values **42** of the first edge device **22-1** of the data structure **40**. The computational requirements **44** of performing the action **20** on the first edge device **22-1** may indicate that the action **20** requires 25% of the battery of the first edge device **22-1**. The charge exchange service **18** can then determine that the battery level **62** of the first edge device **22-1** (i.e., computing resource **50**) does not meet the computational requirements **44** of performing the action **20** on the first edge device **22-1** because the action **20** requires 25% battery but the first edge device **22-1** has 20% battery, and inhibit the action **20** from being performed on the first edge device **22-1** as a result.

(27) In some examples, the data structure **40** may contain rules that identify a safe level of a computing resource (e.g., computing resource **50**, computing resource **52**), where the charge exchange service **18** will not allow the edge device to perform an action that will reduce the computing resource to an amount below the safe level. The safe level may be the lowest level of a computing resource that allows the edge device to remain operational in the computing system or network or a level of a computing resource that keeps the edge device from overheating, as non-limiting examples. For example, the computing resource values **42** of the first edge device **22-1** may indicate that the safe level of battery percentage for the first edge device is 25% and the computational requirements **44** of performing the action **20** on the first edge device **22-1** may indicate that the action **20** requires an amount of battery that will reduce the battery of the first edge device **22-1** to an amount below the 25% safe level of battery percentage for the first edge device **22-1**, so the charge exchange service **18** will not allow the first edge device **22-1** to perform the action. In another example, the computing resource values **42** of the first edge device **22-1** may indicate that the safe level of available memory **56** of the first edge device **22-1** is 1 GB and the computational requirements **44** of performing the action **20** on the first edge device **22-1** may indicate that the action **20** requires an amount of memory that will reduce the amount of available memory **56** to an amount below the 1 GB safe level of memory for the first edge device **22-1**, so

the charge exchange service **18** will not allow the first edge device **22-1** to perform the action **20**.
(28) In another example, subsequent to inhibiting the action **20** from being performed on the first edge device **22-1**, the charge exchange service **18** may determine, based on the data structure **40**, that the first edge device **22-1** has a computing resource (e.g., the computing resource **50**) sufficient to meet the computational requirements **44** of performing the action **20** on the first edge device **22-1**. The charge exchange service **18** may then permit the action **20** to be performed on the first edge device **22-1**. For example, the computing resource **50** of the first edge device **22-1** may be the battery level **62** of the first edge device **22-1** with a value of 90%, which may be identified in the quantity of a computing resource **46** of the first edge device **22-1** in the computing resource values **42** of the first edge device **22-1** of the data structure **40**. The computational requirements **44** of performing the action **20** on the first edge device **22-1** may indicate that the action **20** requires 25% of the battery of the first edge device **22-1**. The charge exchange service **18** can then determine that the battery level **62** of the first edge device **22-1** (i.e., computing resource **50**) meets the computational requirements **44** of performing the action **20** on the first edge device **22-1** because the action **20** requires 25% battery and the first edge device **22-1** has 90% battery, and permit the action **20** to be performed on the first edge device **22-1** as a result. In another example, the charge exchange service **18** may determine, based on the data structure **40**, that the first edge device **22-1** has a computing resource (e.g., the computing resource **50**) sufficient to meet the computational requirements **44** of performing the action **20** on the first edge device **22-1**, and permit the action **20** to be performed on the first edge device **22-1** as a result.

(29) In another example, subsequent to permitting the action **20** to be performed on the first edge device **22-1**, the charge exchange service **18** may determine that the first edge device **22-1** lacks a computing resource (e.g., the computing resource **50**) sufficient to meet the computational requirements **44** of performing the action **20** on the first edge device **22-1**. The charge exchange service may then inhibit the action **20** from being performed on the first edge device **22-1**. For instance, the charge exchange service **18** may determine, based on the data structure **40**, that the first edge device **22-1** has a quantity of the computing resource **50** that is sufficient to meet the computational requirements **44** of performing the action **20** on the first edge device **22-1** and permit the first edge device **22-1** to perform the action **20**, and then a condition in the computing system, network, computing device, or edge device (e.g., the first edge device **22-1**) may occur which changes the quantity of the computing resource **50**. As a result, the charge exchange service **18** may determine that the first edge device **22-1** lacks the computing resource **50** sufficient to meet the computational requirements **44** of performing the action **20** on the first edge device **22-1**, and inhibit the first edge device **22-1** from continuing to perform the action **20** as a result. For example, the computing resource **50** of the first edge device **22-1** may be the battery level **62** of the first edge device **22-1** with a value of 90%, which may be identified in the quantity of a computing resource **46** of the first edge device **22-1** in the computing resource values **42** of the first edge device **22-1** of the data structure **40**. The computational requirements **44** of performing the action **20** on the first edge device **22-1** may indicate that the action **20** requires 25% of the battery of the first edge device **22-1**. The charge exchange service **18** can then determine that the battery level **62** of the first edge device **22-1** (i.e., computing resource **50**) meets the computational requirements **44** of performing the action **20** on the first edge device **22-1** because the action **20** requires 25% battery and the first edge device **22-1** has 90% battery, and permit the action **20** to be performed on the first edge device **22-1** as a result. Then an event may occur before the action **20** is commenced or completed which reduces the battery of the first edge device **22-1** to 20%. The charge exchange service **18** may then pause or inhibit the action **20** from being performed on the first edge device **22-1** because the battery level **62** of the first edge device **22-1** (i.e., computing resource **50**) no longer meets the computational requirements **44** of performing the action **20** on the first edge device **22-1**, as the action **20** requires 25% battery and the first edge device **22-1** has 20% battery.

(30) It is to be understood that, because the charge exchange service **18** is a component of the

computing device **10**, functionality implemented by the charge exchange service **18** may be attributed to the computing device **10** generally. Moreover, in examples where the charge exchange service **18** comprises software instructions that program the processor device **14** to carry out functionality discussed herein, functionality implemented by the charge exchange service **18** may be attributed herein to the processor device **14**. It is to be further understood that while, for purposes of illustration only, the charge exchange service **18** is depicted as a single component, the functionality implemented by the charge exchange service **18** may be implemented in any number of components, and the examples discussed herein are not limited to any particular number of components.

(31) FIG. 2 is a flowchart illustrating operations performed by the computing device of FIG. 1 for minimum viable charge exchange for service operations in edge devices, according to one example. Elements of FIG. 1 are referenced in describing FIG. 2 for the sake of clarity. In FIG. 2, operations begin with a processor device of a computing device, such as the processor device **14** of the computing device **10** of FIG. 1, determining that an action **20** should be performed on a first edge device **22-1** operating in a computing system (block **200**). The processor device **14** then accesses a data structure **40** that identifies values of computing resources **42** of the first edge device **22-1** and computational requirements **44** of performing the action **20** on the first edge device **22-1** (block **202**). The processor device **14** then determines, based on the data structure **40**, that the first edge device **22-1** lacks a computing resource **50** sufficient to meet the computational requirements **44** of performing the action **20** on the first edge device **22-1** (block **204**). The processor device **14** then, in response to determining that the first edge device **22-1** lacks the computing resource **50** sufficient to meet the computational requirements **44** of performing the action **20** on the first edge device **22-1**, inhibits the action **20** from being performed on the first edge device **22-1** (block **206**).

(32) FIG. 3 is a block diagram of the computing device of FIG. 1 for minimum viable charge exchange for service operations in edge devices, according to one example. Elements of FIG. 1 are referenced in describing FIG. 3 for the sake of clarity. In the example of FIG. 3, the charge exchange service **18** may, subsequent to inhibiting the action **20** from being performed on the first edge device **22-1**, obtain the computing resource values **42** of the first edge device **22-1** from the data structure **40**. The computing resource values **42** of the first edge device **22-1** may include a battery level **82** of the first edge device **22-1** and a threshold battery level **84** for the first edge device **22-1**. In some examples, the battery level **82** of the first edge device **22-1** may be identified in the quantity of a computing resource **46** of the first edge device **22-1** and the threshold battery level **84** for the first edge device may be identified in the quantity of a computing resource **48** of the first edge device **22-1**, both the quantity of a computing resource **46** of the first edge device **22-1** and the quantity of a computing resource **48** of the first edge device **22-1** being identified in the computing resource values **42** of the first edge device **22-1** of the data structure **40**. The battery level **82** of the first edge device **22-1** may correlate to the computing resource **50** of the first edge device **22-1**, where the computing resource **50** of the first edge device **22-1** is the battery level **62-1** of the first edge device **22-1**.

(33) The charge exchange service **18** may determine that the battery level **82** of the first edge device **22-1** is below the threshold battery level **84** for the first edge device **22-1**. For example, the charge exchange service **18** may obtain the battery level **82** of the first edge device **22-1** from the computing resource values **42** of the first edge device **22-1** that indicates that the battery level **62-1** of the first edge device is 10% and obtain the threshold battery level **84** for the first edge device **22-1** from the computing resource values **42** of the first edge device **22-1** that indicates that the threshold battery level **84** for the first edge device **22-1** is 20%. Then the charge exchange service **18** can determine that the 10% battery level **62-1** of the first edge device **22-1** is below the 20% threshold battery level **84** for the first edge device **22-1**.

(34) The charge exchange service **18** may obtain computing resource values **86** of a second edge device **22-2** from the data structure **40**. The computing resource values **86** of the second edge

device **22-2** may include a battery level **88** of the second edge device **22-2** and a threshold battery level **90** for the second edge device **22-2**. In some examples, the battery level **88** of the second edge device **22-2** may be identified in a first quantity of a computing resource of the second edge device and the threshold battery level **90** for the second edge device **22-2** may be identified in a second quantity of a computing resource of the second edge device, both the first quantity of a computing resource of the second edge device and the second quantity of a computing resource of the second edge device being identified in the computing resource values **86** of the second edge device of the data structure **40**. The battery level **88** of the second edge device **22-2** may correlate to a computing resource of the second edge device **22-2**, where the computing resource of the second edge device **22-2** is the battery level **62-2** of the second edge device **22-2**.

(35) The charge exchange service **18** may determine that the battery level **88** of the second edge device **22-2** is above the threshold battery level **90** for the second edge device **22-2**. For example, the charge exchange service **18** may obtain the battery level **88** of the second edge device **22-2** from the computing resource values **86** of the second edge device **22-2** that indicates that the battery level **62-2** of the second edge device is 90% and obtain the threshold battery level **90** for the second edge device **22-2** from the computing resource values **86** of the second edge device **22-2** that indicates that the threshold battery level **90** for the second edge device **22-2** is 20%. Then the charge exchange service **18** can determine that the 90% battery level **62-2** of the second edge device **22-2** is above the 20% threshold battery level **90** for the second edge device **22-2**.

(36) The charge exchange service **18** may then cause the second edge device **22-2** to share energy from a battery **92-2** of the second edge device **22-2** with a battery **92-1** of the first edge device **22-1**. For example, the charge exchange service **18** may obtain the computing resource values **42** of the first edge device **22-1** that identify the threshold battery level **84** for the first edge device **22-1** as 20% and the battery level **82** of the first edge device **22-1**, which correlates to the battery level **62-1** of the first edge device **22-1**, as 10%. The charge exchange service **18** may obtain the computing resource values **86** of the second edge device **22-2** that identify the threshold battery level **90** for the second edge device **22-2** as 20% and the battery level **88** of the second edge device **22-2**, which correlates to the battery level **62-2** of the second edge device **22-2**, as 90%. The charge exchange service **18** may determine that the 10% battery level **62-1** of the first edge device is below the 20% threshold battery level **84** for the first edge device **22-1** and the 90% battery level **62-2** of the second edge device **22-2** is above the 20% threshold battery level **90** for the second edge device **22-2**. The charge exchange service **18** can determine that the second edge device **22-2** has a battery level of 70% available to transfer to another edge device in the computing system or network while remaining able to operate in the computing system or network because the threshold battery level **90** is 20% and the battery level **62-2** is 90%. The charge exchange service **18** can determine that the first edge device **22-1** requires at least 10% more battery level in order to reach the threshold battery level **84** because the battery level **62-1** is 10% and the threshold battery level **84** is 20%. The charge exchange service **18** can cause the second edge device **22-2** to share or transfer energy from the battery **92-2** of the second edge device **22-2** in an amount in the range of 10%-70% in order to bring the 10% battery level **62-1** of the first edge device **22-1** to a level at or above the 20% threshold battery level **84** for the first edge device **22-1** while keeping the battery level **88** for the second edge device **22-2** at or above the 20% threshold battery level **90** for the second edge device **22-2**.

(37) FIG. **4** is a block diagram of the computing device of FIG. **1** for minimum viable charge exchange for service operations in edge devices, according to one example. Elements of FIG. **1** are referenced in describing FIG. **4** for the sake of clarity. In the example of FIG. **4**, the charge exchange service **18** may receive a signal from a server **94** in order to determine that the action **20** should be performed on the first edge device **22-1**. The signal may indicate that one or more of a new application **96**, new firmware **98**, new operating system **100**, new package **102**, new service **104**, or new software **106**, as non-limiting examples, are available for the first edge device **22-1** to

install. The signal may indicate that one or more of a new version of an application, firmware, operating system, package, service, or software on the first edge device **22-1**, as non-limiting examples, are available for the first edge device **22-1** to update. After determining that the action **20** should be performed on the first edge device **22-1**, the charge exchange service **18** may determine the computing resource values **42** of the first edge device **22-1** by polling the first edge device **22-1**. The charge exchange service **18** may then update the data structure **40** with the computing resource values **42** of the first edge device **22-1** that were obtained by polling the first edge device **22-1**. For example, the charge exchange service **18** may receive a signal from the server **94** that indicates that new software **106** is available to install on the first edge device **22-1**. The charge exchange service **18** may determine that the action **20** to be performed is to install the new software **106** on the first edge device **22-1**. The charge exchange service **18** can then poll the first edge device **22-1** to determine the value of a computing resource **50** on the first edge device **22-1**, such as the battery level of the first edge device **22-1**. The charge exchange service **18** can update the computing resource values **42** of the first edge device **22-1** in the data structure **40** with the value of the computing resource **50** (e.g., the battery level) of the first edge device **22-1** that was obtained by polling the first edge device **22-1**.

(38) In some examples, the data structure **40** may be updated periodically (e.g., hourly or daily) with updated values for the computing resource values of the first edge device **22-1** and for the computational requirements **44** of performing the action **20** on the first edge device **22-1**. The data structure **40** may also be updated periodically (e.g., hourly or daily) with updated computing resource values of other edge devices in the computing system or network and with updated computational requirements of performing actions on the edge devices in the computing system or network. In other examples, the data structure **40** may be updated when an edge device in the computing system or network (e.g., the first edge device **22-1**) sends a notification to the charge exchange service **18** indicating that the battery level of the edge device is at the level of a predefined threshold, then the charge exchange service **18** can update the data structure **40** with the battery level of the edge device from the notification in the computing resource values (e.g., computing resource values **42**, computing resource values **78**) for the edge device. For example, edge devices in the computing system or network may be configured to send a notification to the charge exchange service **18** when the battery level of the edge device reaches 15%, then the charge exchange service **18** can update the computing resource values (e.g., computing resource values **42**, computing resource values **78**) for the edge device in the data structure **40** to indicate a 15% battery level when an edge device has 15% battery and sends a notification to the charge exchange service **18** that specifies the edge device has 15% battery.

(39) In another example, an edge device in the computing system or network can send a signal to the charge exchange service **18** when the edge device reaches a predetermined battery threshold in order to determine whether any updates to the components of the edge device are available to install. For example, when an edge device reaches an 80% battery level, the edge device may send a signal to the charge exchange service **18** requesting an update to or a new version of an application, firmware, operating system, package, service, or software, as non-limiting examples, and the charge exchange service **18** may request the update or new version from the server **94**. If the server **94** sends, and the charge exchange service **18** receives, a signal that a new version or update is available, then the new version or update can be preloaded or executed on the edge device.

(40) FIG. 5 is a block diagram of the computing device of FIG. 1 for minimum viable charge exchange for service operations in edge devices, according to one example. Elements of FIG. 1 are referenced in describing FIG. 5 for the sake of clarity. In the example of FIG. 5, the charge exchange service **18** may obtain the computing resource values **42** of the first edge device **22-1** from the data structure **40**. The computing resource values **42** of the first edge device **22-1** may include the threshold battery level **84** for the first edge device **22-1**. The threshold battery level **84**

for the first edge device **22-1** can indicate a minimum viable battery charge **108** for the first edge device **22-1**. The minimum viable battery charge **108** for the first edge device **22-1** may identify the lowest amount of battery charge of the first edge device **22-1** that allows the first edge device **22-1** to operate in the computing system or network. The charge exchange service **18** may obtain the computational requirements **44** of performing the action **20** on the first edge device **22-1** from the data structure **40**. The computational requirements **44** of performing the action **20** on the first edge device **22-1** may include the amount of battery **74** required to perform the action **20** on the first edge device **22-1**. The charge exchange service **18** can determine that the amount of battery **74** required to perform the action **20** exceeds the threshold battery level **84** for the first edge device **22-1**. The charge exchange service **18** may determine that the first edge device **22-1** lacks the computing resource **50** (e.g., the battery level of the first edge device **22-1**) sufficient to meet the computational requirements **44** of performing the action **20** on the first edge device **22-1** if the charge exchange service **18** determines that the amount of battery **74** required to perform the action **20** exceeds the threshold battery level **84** for the first edge device **22-1**.

(41) For example, the charge exchange service **18** may obtain from the data structure **40** the computing resource values **42** of the first edge device **22-1** that indicates that the threshold battery level **84** for the first edge device **22-1** is 10%, where 10% is the minimum viable battery charge **108** for the first edge device **22-1** to be able to operate in the computing system or network. The charge exchange service **18** may obtain from the data structure **40** the computational requirements **44** of performing the action **20** on the first edge device **22-1** that indicates that performing the action **20** requires 15% of the battery of the first edge device **22-1**. The computing resource **50** of the first edge device **22-1** may indicate that the battery level of the first edge device **22-1** is 20%. The charge exchange service **18** can then determine that the amount of battery **74** required to perform the action **20** exceeds the threshold battery level **84** for the first edge device **22-1** because the battery level of the first edge device **22-1** is 20%, the amount of battery **74** required to perform the action **20** is 15%, and the threshold battery level **84** for the first edge device **22-1** is 10%, so the action **20** cannot be performed without the battery level of the first edge device **22-1** falling below the threshold battery level **84** for the first edge device **22-1**, which would leave the first edge device **22-1** inoperable in the computing system or network. The charge exchange service **18** can then determine that the first edge device **22-1** lacks the computing resource **50** (the battery level of the first edge device **22-1**) sufficient to meet the computational requirements **44** of performing the action **20** on the first edge device **22-1**.

(42) FIG. **6** is a block diagram of the computing device of FIG. **1** for minimum viable charge exchange for service operations in edge devices, according to one example. Elements of FIG. **1** are referenced in describing FIG. **6** for the sake of clarity. In the example of FIG. **6**, the charge exchange service **18** may obtain the computing resource values **42** of the first edge device **22-1** from the data structure **40**. The computing resource values **42** of the first edge device **22-1** may include the threshold battery level **84** for the first edge device **22-1**. The threshold battery level **84** for the first edge device **22-1** may identify the minimum viable battery charge **108** for the first edge device **22-1**. The charge exchange service **18** may obtain the computational requirements **44** of performing the action **20** on the first edge device **22-1** from the data structure **40**. The computational requirements **44** of performing the action **20** on the first edge device **22-1** may include the amount of battery **74** required to perform the action **20** on the first edge device **22-1** and an amount of battery required to reverse **110** the action **20** on the first edge device **22-1**. The amount of battery required to reverse **110** the action **20** on the first edge device **22-1** may include the amount of battery required to roll back the action **20** and to reassert an older version of the component, such as the application, firmware, operating system, service, or software, to which the action **20** correlates. In other examples, the computing resource values **42** of the first edge device **22-1** may include quantities of other computing resources (e.g., computing resource **50**, computing resource **52**) on the first edge device **22-1**, such as available memory **56** or available storage **58**,

and the computational requirements **44** of performing the action **20** on the first edge device **22-1** may include the amount of computing resources, such as memory **68** or storage **70**, required to perform the action **20** on the first edge device **22-1** and to reverse the action **20** on the first edge device **22-1**. The amount of the computing resources required to reverse the action **20** on the first edge device **22-1** may include the amount of computing resources required to roll back the action **20** and to reassert an older version of the component, such as the application, firmware, operating system, service, or software, to which the action **20** correlates.

(43) The charge exchange service **18** may determine that the amount of battery **74** required to perform the action **20** on the first edge device **22-1** together with the amount of battery required to reverse **110** the action **20** on the first edge device **22-1** exceeds the threshold battery level **84** for the first edge device **22-1**. The charge exchange service **18** may determine that the first edge device **22-1** lacks the computing resource **50** (e.g., the battery level of the first edge device **22-1**) sufficient to meet the computational requirements **44** of performing the action **20** on the first edge device **22-1** if the charge exchange service **18** determines that the amount of battery **74** required to perform the action **20** on the first edge device **22-1** together with the amount of battery required to reverse **110** the action **20** on the first edge device **22-1** exceeds the threshold battery level **84** for the first edge device **22-1**. As a result, the charge exchange service **18** can pause the action **20** or inhibit the action **20** from being performed on the first edge device **22-1**.

(44) For example, the charge exchange service **18** may obtain from the data structure **40** the computing resource values **42** of the first edge device **22-1** that indicates that the threshold battery level **84** for the first edge device **22-1** is 20%, where 20% is the minimum viable battery charge **108** for the first edge device **22-1** to be able to operate in the computing system or network. The charge exchange service **18** may obtain from the data structure **40** the computational requirements **44** of performing the action **20** on the first edge device **22-1** that indicates that the amount of battery **74** required to perform the action **20** on the first edge device **22-1** is 5% and the amount of battery required to reverse **110** the action **20** on the first edge device **22-1** is 10%. The computing resource **50** of the first edge device **22-1** may indicate that the battery level of the first edge device **22-1** is 25%. The charge exchange service **18** can then determine that the amount of battery **74** required to perform the action **20** together with the amount of battery required to reverse **110** the action **20** on the first edge device **22-1** exceeds the threshold battery level **84** for the first edge device **22-1** because the battery level of the first edge device **22-1** is 25%, the amount of battery **74** required to perform the action **20** is 5%, the amount of battery required to reverse **110** the action **20** is 10%, and the threshold battery level **84** for the first edge device **22-1** is 20%, so the action **20** cannot be performed and reversed, if necessary, without the battery level of the first edge device **22-1** falling below the threshold battery level **84** for the first edge device **22-1**, which would leave the first edge device **22-1** inoperable in the computing system or network. The charge exchange service **18** can then determine that the first edge device **22-1** lacks the computing resource **50** (the battery level of the first edge device **22-1**) sufficient to meet the computational requirements **44** of performing the action **20** on the first edge device **22-1**.

(45) FIG. 7 is a block diagram of the computing device of FIG. 1 for minimum viable charge exchange for service operations in edge devices, according to one example. Elements of FIG. 1 are referenced in describing FIG. 7 for the sake of clarity. In the example of FIG. 7, the charge exchange service **18** may obtain the computing resource values **42** of the first edge device **22-1** from the data structure **40**. The computing resource values **42** of the first edge device **22-1** may contain multiple entries that correlate to the quantities of multiple computing resources (e.g., computing resource **50**, computing resource **52**) of the first edge device **22-1**. The computing resource values **42** of the first edge device **22-1** may include the battery level **82** of the first edge device **22-1** and an age **112** of the first edge device **22-1**. For instance, the battery level **82** may correlate to the computing resource **50** of the first edge device **22-1** and the age **112** may correlate to the computing resource **52** of the first edge device **22-1**. The charge exchange service **18** may

obtain the computing resource values **86** of the second edge device **22-2** from the data structure **40**. The computing resource values **86** of the second edge device **22-2** may contain multiple entries that correlate to the quantities of multiple computing resources, such as a computing resource **118** and a computing resource **120** of the second edge device **22-2**. The computing resource values **86** of the second edge device **22-2** may include a battery level **114** of the second edge device **22-2** and an age **116** of the second edge device **22-2**. For instance, the battery level **114** may correlate to the computing resource **118** of the second edge device **22-2** and the age **116** may correlate to the computing resource **120** of the second edge device **22-2**. The charge exchange service **18** can determine that the battery level **82** of the first edge device **22-1** decreases at a faster rate than the battery level **114** of the second edge device **22-2**, based on the age **112** of the first edge device **22-1** and the age **116** of the second edge device **22-2**. For instance, the age **112** of the first edge device **22-1** may be greater than the age **116** of the second edge device **22-2** and, as a result, the battery level **82** of the first edge device **22-1** may decrease faster than the battery level **114** of the second edge device **22-2**. The charge exchange service **18** can determine that the first edge device **22-1** lacks the computing resource **50** (e.g., the battery level of the first edge device **22-1**) sufficient to meet the computational requirements **44** of performing the action on the first edge device **22-1** if the battery level **82** of the first edge device **22-1** decreases at a faster rate than the battery level **114** of the second edge device **22-2**.

(46) FIG. **8** is a block diagram of the computing device **10** of FIG. **1** for minimum viable charge exchange for service operations in edge devices, according to one example. Elements of FIG. **1** are referenced in describing FIG. **8** for the sake of clarity. In the example of FIG. **8**, the computing device **10** includes the charge exchange service **18**, a system memory **12**, and a processor device **14** coupled to the system memory **12**. The processor device **14** is to determine that an action **20** should be performed on a first edge device **22-1** operating in a computing system. The processor device **14** is further to access a data structure **40** that identifies values of computing resources **42** of the first edge device **22-1** and computational requirements **44** of performing the action **20** on the first edge device **22-1**. The processor device **14** is further to determine, based on the data structure **40**, that the first edge device **22-1** lacks a computing resource **50** to meet the computational requirements **44** of performing the action **20** on the first edge device **22-1**. The processor device **14** is further to, in response to determining that the first edge device **22-1** lacks the computing resource **50** to meet the computational requirements **44** of performing the action **20** on the first edge device **22-1**, inhibit the action **20** from being performed on the first edge device **22-1**.

(47) FIG. **9** is a block diagram of the computing device **10** suitable for implementing examples according to one example. The computing device **10** may comprise any computing or electronic device capable of including firmware, hardware, and/or executing software instructions to implement the functionality described herein. The computing device **10** includes the processor device **14**, the system memory **12**, and a system bus **122**. The system bus **122** provides an interface for system components including, but not limited to, the system memory **12** and the processor device **14**. The processor device **14** can be any commercially available or proprietary processor.

(48) The system bus **122** may be any of several types of bus structures that may further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and/or a local bus using any of a variety of commercially available bus architectures. The system memory **12** may include non-volatile memory **124** (e.g., read-only memory (ROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), etc.), and volatile memory **126** (e.g., random-access memory (RAM)). A basic input/output system (BIOS) **128** may be stored in the non-volatile memory **124** and can include the basic routines that help to transfer information between elements within the computing device **10**. The volatile memory **126** may also include a high-speed RAM, such as static RAM, for caching data.

(49) The computing device **10** may further include or be coupled to a non-transitory computer-readable storage medium such as a storage device **16**, which may comprise, for example, an

internal or external hard disk drive (HDD) (e.g., enhanced integrated drive electronics (EIDE) or serial advanced technology attachment (SATA)), HDD (e.g., EIDE or SATA) for storage, flash memory, or the like. The storage device **16** and other drives associated with computer-readable media and computer-usable media may provide non-volatile storage of data, data structures, computer-executable instructions, and the like.

(50) A number of modules can be stored in the storage device **16** and in the volatile memory **126**, including an operating system **130** and one or more program modules, such as the charge exchange service **18**, which may implement the functionality described herein in whole or in part. All or a portion of the examples may be implemented as a computer program product **132** stored on a transitory or non-transitory computer-usable or computer-readable storage medium, such as the storage device **16**, which includes complex programming instructions, such as complex computer-readable program code, to cause the processor device **14** to carry out the steps described herein. Thus, the computer-readable program code can comprise software instructions for implementing the functionality of the examples described herein when executed on the processor device **14**. The processor device **14**, in conjunction with the charge exchange service **18** in the volatile memory **126**, may serve as a controller, or control system, for the computing device **10** that is to implement the functionality described herein.

(51) An operator, such as a user, may also be able to enter one or more configuration commands through a keyboard (not illustrated), a pointing device such as a mouse (not illustrated), or a touch-sensitive surface such as a display device (not illustrated). Such input devices may be connected to the processor device **14** through an input device interface **134** that is coupled to the system bus **122** but can be connected by other interfaces such as a parallel port, an Institute of Electrical and Electronic Engineers (IEEE) 1394 serial port, a Universal Serial Bus (USB) port, an IR interface, and the like. The computing device **10** may also include a communications interface **136** suitable for communicating with the network as appropriate or desired. The computing device **10** may also include a video port (not illustrated) configured to interface with the display device (not illustrated), to provide information to the user.

(52) Individuals will recognize improvements and modifications to the preferred examples of the disclosure. All such improvements and modifications are considered within the scope of the concepts disclosed herein and the claims that follow.

Claims

1. A method, comprising: determining, by a computing device, that an action should be performed on a first edge device operating in a computing system; accessing, by the computing device, a data structure that identifies values of computing resources of the first edge device and computational requirements of performing the action on the first edge device, wherein the computational requirements of performing the action comprise an amount of battery required to perform the action, and wherein the values of the computing resources of the first edge device comprise a current battery level of the first edge device; determining, by the computing device, that a difference between the current battery level and the amount of battery required to perform the action is less than a minimum threshold battery level for the first edge device; determining, by the computing device based on the data structure, that the first edge device lacks a computing resource to meet the computational requirements of performing the action on the first edge device; and in response to determining that the first edge device lacks the computing resource to meet the computational requirements of performing the action on the first edge device, inhibiting, by the computing device, the action from being performed on the first edge device.

2. The method of claim 1, further comprising: subsequent to inhibiting the action from being performed on the first edge device, determining, by the computing device based on the data structure, that the first edge device has the computing resource to meet the computational

requirements of performing the action on the first edge device; and in response to determining that the first edge device has the computing resource to meet the computational requirements of performing the action on the first edge device, permitting, by the computing device, the action to be performed on the first edge device.

3. The method of claim 2, further comprising: subsequent to permitting the action to be performed on the first edge device, determining, by the computing device, that the first edge device lacks the computing resource to meet the computational requirements of performing the action on the first edge device; and in response to determining that the first edge device lacks the computing resource to meet the computational requirements of performing the action on the first edge device, inhibiting, by the computing device, the action from being performed on the first edge device.

4. The method of claim 1, further comprising: subsequent to inhibiting the action from being performed on the first edge device, obtaining, from the data structure, the values of the computing resources of the first edge device that include an updated battery level of the first edge device; determining that a difference between the updated battery level of the first edge device and the amount of battery required to perform the action is below the minimum threshold battery level for the first edge device; obtaining, from the data structure, values of computing resources of a second edge device that include a current battery level of the second edge device and a minimum threshold battery level for the second edge device; determining that the current battery level of the second edge device is above the minimum threshold battery level for the second edge device; and causing, by the computing device, the second edge device to share energy from a battery of the second edge device with a battery of the first edge device.

5. The method of claim 1, wherein the action comprises updating one or more of an application, firmware, an operating system, a package, a service, or software on the first edge device.

6. The method of claim 1, wherein the action comprises installing one or more of software and a service on the first edge device.

7. The method of claim 1, wherein the data structure identifies values of computing resources of a plurality of edge devices operating in the computing system and computational requirements of performing actions on the plurality of edge devices.

8. The method of claim 1, wherein the values of the computing resources of the first edge device further comprise quantities of the computing resources of the first edge device.

9. The method of claim 8, wherein the computing resources of the first edge device comprise one or more of CPU cycles, memory available, storage available, load, battery level, and age of the first edge device.

10. The method of claim 1, wherein the computational requirements of performing the action on the first edge device further comprise one or more of CPU cycles, memory, storage, network capabilities, battery level, and time required to perform the action on the first edge device.

11. The method of claim 1, wherein determining that the action should be performed on the first edge device comprises receiving a signal from a server indicating that a new version of one or more of an application, firmware, an operating system, a package, a service, or software is available for the first edge device.

12. The method of claim 1, further comprising: subsequent to determining that the action should be performed on the first edge device, determining, by the computing device, the values of the computing resources of the first edge device by polling the first edge device; and updating, by the computing device, the data structure with the values of the computing resources of the first edge device from polling the first edge device.

13. The method of claim 1, wherein determining, based on the data structure, that the first edge device lacks the computing resource to meet the computational requirements of performing the action on the first edge device comprises: obtaining, from the data structure, the values of the computing resources of the first edge device that includes the minimum threshold battery level for the first edge device; and obtaining, from the data structure, the computational requirements of

performing the action on the first edge device that includes the amount of battery required to perform the action.

14. The method of claim 1, wherein the minimum threshold battery level for the first edge device comprises a minimum viable battery charge for the first edge device to operate in the computing system.

15. The method of claim 1, wherein determining, based on the data structure, that the first edge device lacks the computing resource to meet the computational requirements of performing the action on the first edge device comprises: obtaining, from the data structure, the values of the computing resources of the first edge device that include the minimum threshold battery level for the first edge device; obtaining, from the data structure, the computational requirements of performing the action on the first edge device that include the amount of battery required to perform the action and an amount of battery required to reverse the action; and determining that the amount of battery required to perform the action together with the amount of battery required to reverse the action exceeds the minimum threshold battery level for the first edge device.

16. The method of claim 1, wherein determining, based on the data structure, that the first edge device lacks the computing resource to meet the computational requirements of performing the action on the first edge device comprises: obtaining, from the data structure, the values of the computing resources of the first edge device that include an age of the first edge device and the current battery level of the first edge device; obtaining, from the data structure, values of computing resources of a second edge device operating in the computing system that includes an age of the second edge device and a current battery level of the second edge device; and determining, based on the age of the first edge device and the age of the second edge device, that a battery charge of the first edge device decreases at a faster rate than a battery charge of the second edge device.

17. A computing device, comprising: a memory; and a processor device coupled to the memory, the processor device to: determine that an action should be performed on a first edge device operating in a computing system; access a data structure that identifies values of computing resources of the first edge device and computational requirements of performing the action on the first edge device, wherein the computational requirements of performing the action comprise an amount of battery required to perform the action, and wherein the values of the computing resources of the first edge device comprise a current battery level of the first edge device; determine that a difference between the current battery level and the amount of battery required to perform the action is less than a minimum threshold battery level for the first edge device; determine, based on the data structure, that the first edge device lacks a computing resource to meet the computational requirements of performing the action on the first edge device; and in response to determining that the first edge device lacks the computing resource to meet the computational requirements of performing the action on the first edge device, inhibit the action from being performed on the first edge device.

18. The computing device of claim 17, wherein the processor device is further to: subsequent to inhibiting the action from being performed on the first edge device, determine, based on the data structure, that the first edge device has the computing resource to meet the computational requirements of performing the action on the first edge device; and in response to determining that the first edge device has the computing resource to meet the computational requirements of performing the action on the first edge device, permit the action to be performed on the first edge device.

19. The computing device of claim 17, wherein the processor device is further to: subsequent to inhibiting the action from being performed on the first edge device, obtain, from the data structure, the values of the computing resources of the first edge device that include an updated battery level of the first edge device; determine that the updated battery level of the first edge device is below the minimum threshold battery level for the first edge device; obtain, from the data structure, values of computing resources of a second edge device that include a current battery level of the

second edge device and a minimum threshold battery level for the second edge device; determine that the current battery level of the second edge device is above the minimum threshold battery level for the second edge device; and cause the second edge device to share energy from a battery of the second edge device with a battery of the first edge device.

20. A non-transitory computer-readable storage medium that includes computer-executable instructions that, when executed, cause one or more processor devices to: determine that an action should be performed on a first edge device operating in a computing system; access a data structure that identifies values of computing resources of the first edge device and computational requirements of performing the action on the first edge device, wherein the computational requirements of performing the action comprise an amount of battery required to perform the action, and wherein the values of the computing resources of the first edge device comprise a current battery level of the first edge device; determine that a difference between the current battery level and the amount of battery required to perform the action is less than a minimum threshold battery level for the first edge device; determine, based on the data structure, that the first edge device lacks a computing resource to meet the computational requirements of performing the action on the first edge device; and in response to determining that the first edge device lacks the computing resource to meet the computational requirements of performing the action on the first edge device, inhibit the action from being performed on the first edge device.
